



NexHR

Intelligent Enterprise Management Solution

 React 18

 Django 5.2

 AI Powered

 Cloud Native

| Project Introduction & Technology Stack

NexHR offers a unified platform for managing employees, payroll, attendance, hiring, and financial operations. It's designed for scalability and efficiency, catering to modern HR needs.

- ✓ **Scalability:** Microservices-ready architecture.
- ✓ **Efficiency:** Automating repetitive tasks with AI.
- ✓ **Security:** Enterprise-grade Role-Based Access Control.

Frontend Stack

React 18

TypeScript

Tailwind CSS

Redux Toolkit

Vite

Backend Stack

Django REST

PostgreSQL

Celery

Redis

pgvector

High-Level Architecture



Client Layer

A responsive Single Page Application (SPA) built with React. Handles routing, state management, and role-based UI rendering.



Service Layer

Django REST API acting as the central nervous system. Handles authentication, business logic, and data validation.



Async Workers

Celery workers manage heavy lifting: AI model inference, PDF generation, and email dispatching via Redis message broker.

| Core Modules



HR Management

Recruitment, Onboarding & Teams



Finance & Payroll

Salary, Tax, & Expense Tracking



Employee Self-Service

Leave, Payslips & Profile



Job Portal

Public Listings & Applications

| HR Management

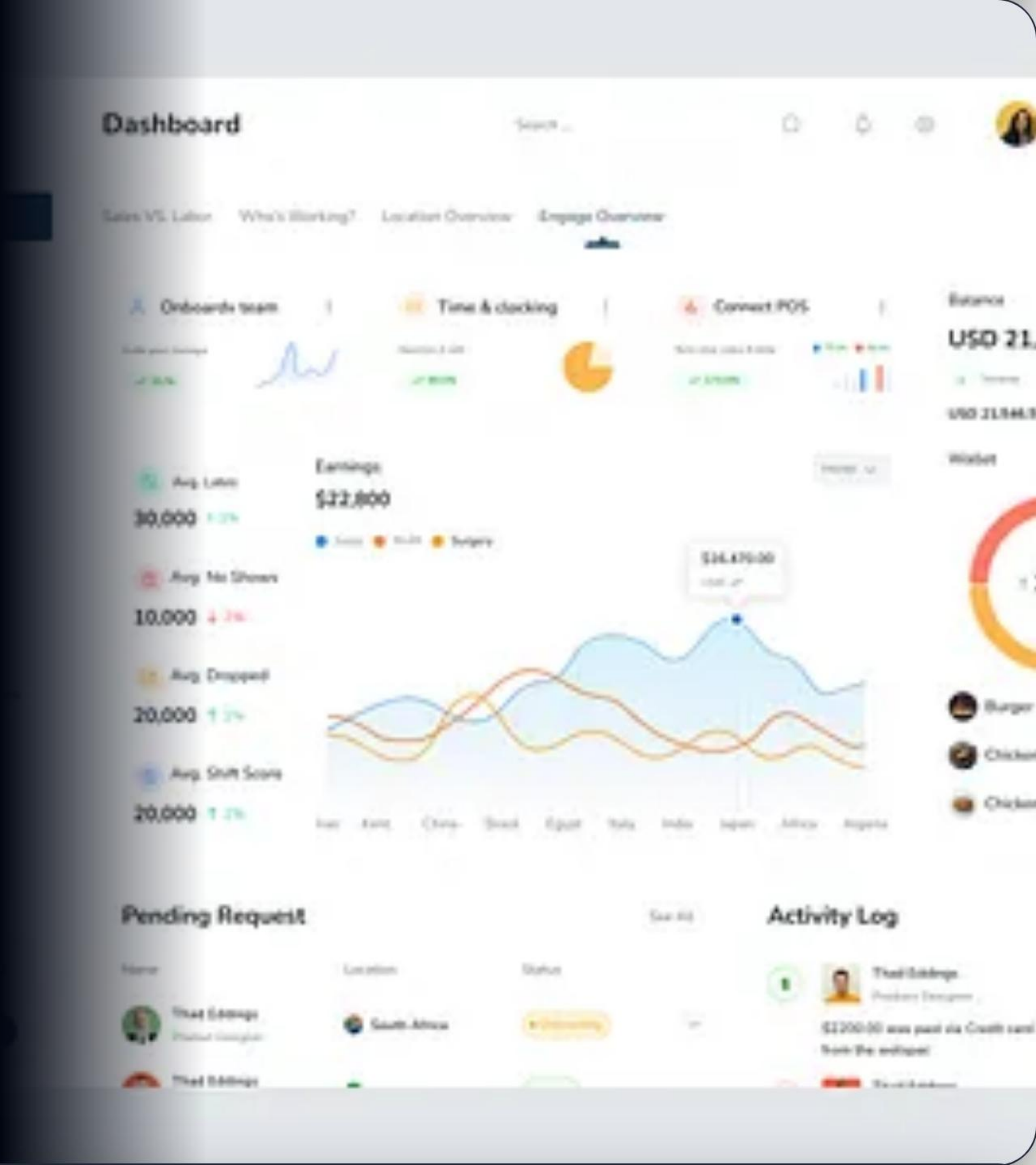
A command center for Talent Acquisition and Workforce Management.

- ✓ **Smart Recruitment:** Track applicants through custom hiring pipelines.
- ✓ **Team Analytics:** Visualize department distribution and growth.
- ✓ **Onboarding:** Automated workflows for new hires.

Finance & Payroll

Automate complex financial operations with precision.

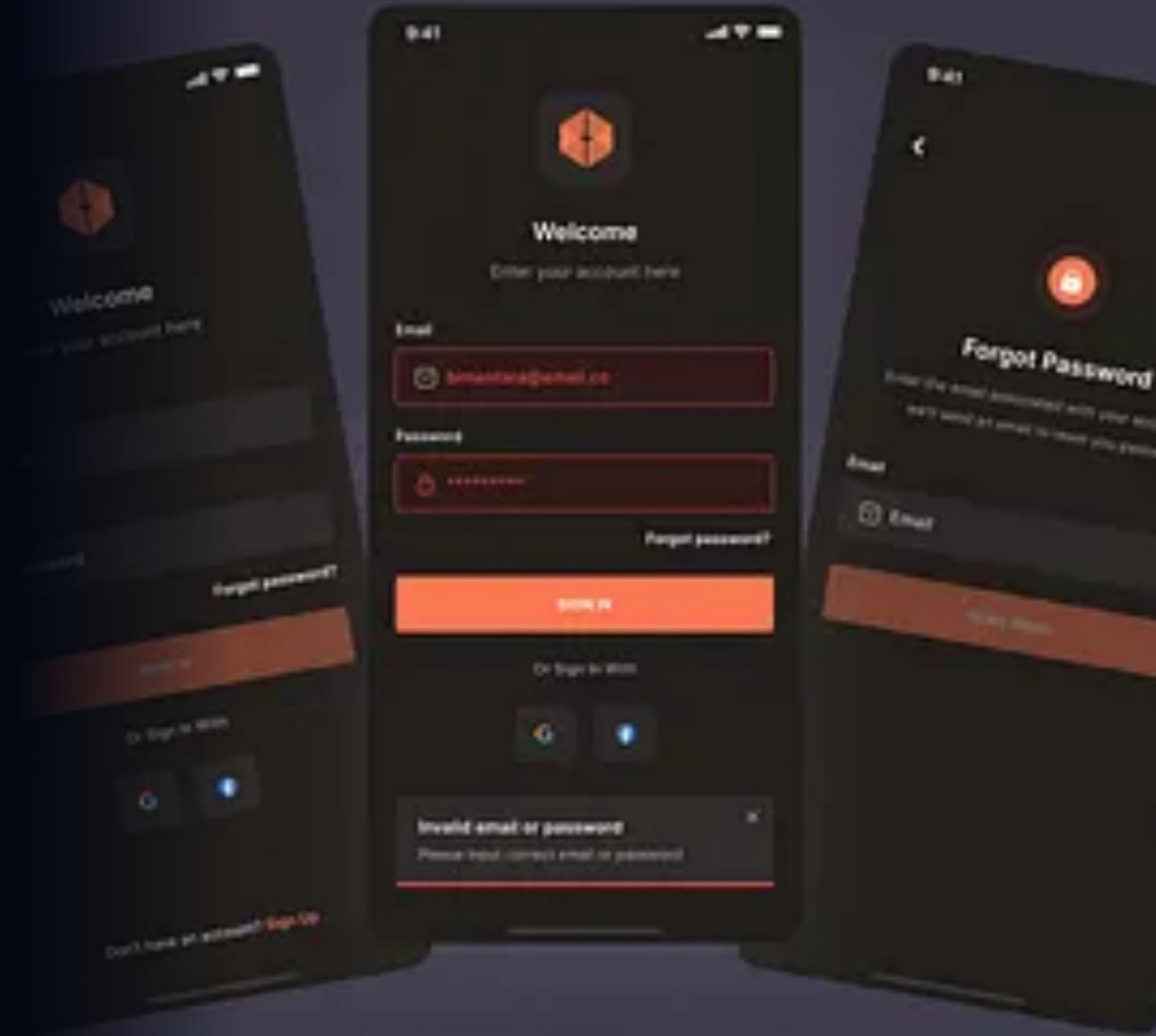
- ✓ **One-Click Payroll:** Calculate and disburse salaries in bulk via Stripe.
- ✓ **Compliance:** Auto-deduction of taxes and statutory components.
- ✓ **Expense Claims:** Streamlined approval workflow for employee reimbursements.



| Employee Portal

Empowering employees with direct access to their data.

- ✓ **Attendance:** View daily logs and shift timings.
- ✓ **Documents:** Download payslips and tax forms instantly.
- ✓ **Requests:** Apply for leave or loans with real-time status tracking.



| Public Job Portal

Your company's public face for attracting top talent.

- ✓ **Branded Listings:** Customizable job posts with rich text descriptions.
- ✓ **Seamless Apply:** Easy application forms with resume parsing.
- ✓ **LinkedIn Sync:** Automatically push jobs to social platforms.

| Secure Authentication Flow



JWT Strategy

Access Tokens: Short-lived, stored in memory for API access.

Refresh Tokens: HttpOnly cookies for secure, silent session renewal.



OAuth 2.0

Integrated with Google and LinkedIn for one-click enterprise login. Eliminates password fatigue and leverages existing identity providers.

| Role-Based Access Control (RBAC)



Admin

Full System Access



HR Manager

Hiring & Employees



Finance

Payroll & Accounts



Employee

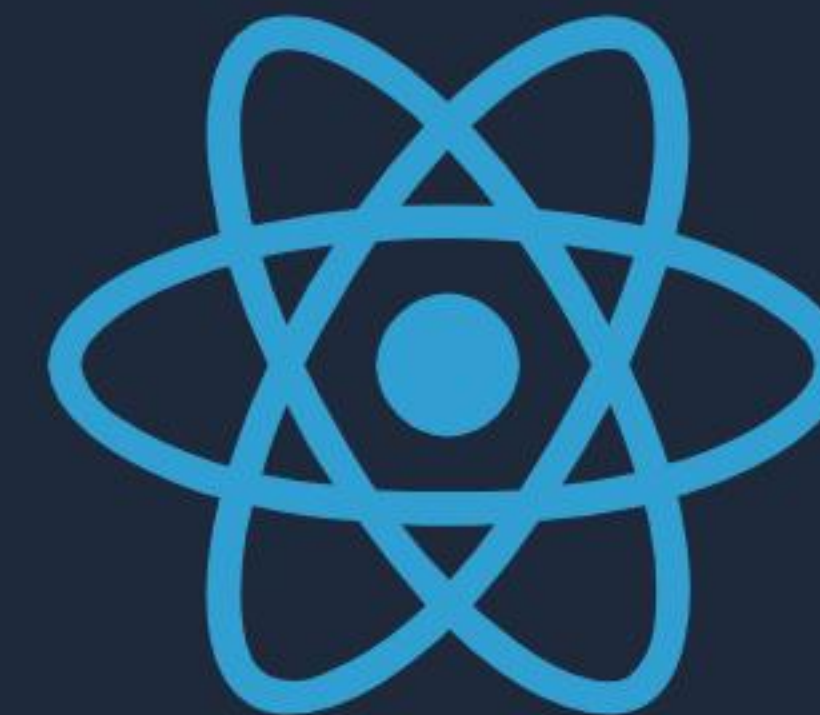
Self Data Only

Dynamic dashboard rendering ensures users only see what they are authorized to access.

| Frontend Component Architecture

Built with **Atomic Design principles** to ensure consistency and reusability.

-  **Atoms:** Buttons, Inputs, Icons (Radix UI)
-  **Molecules:** Form Groups, Search Bars
-  **Organisms:** Data Tables, Sidebar, Charts



| AI Recruitment Engine

Vector Search Technology

Utilizes **pgvector** and **BAI Embeddings** to semantically match candidates to job descriptions.

Automated Scoring:

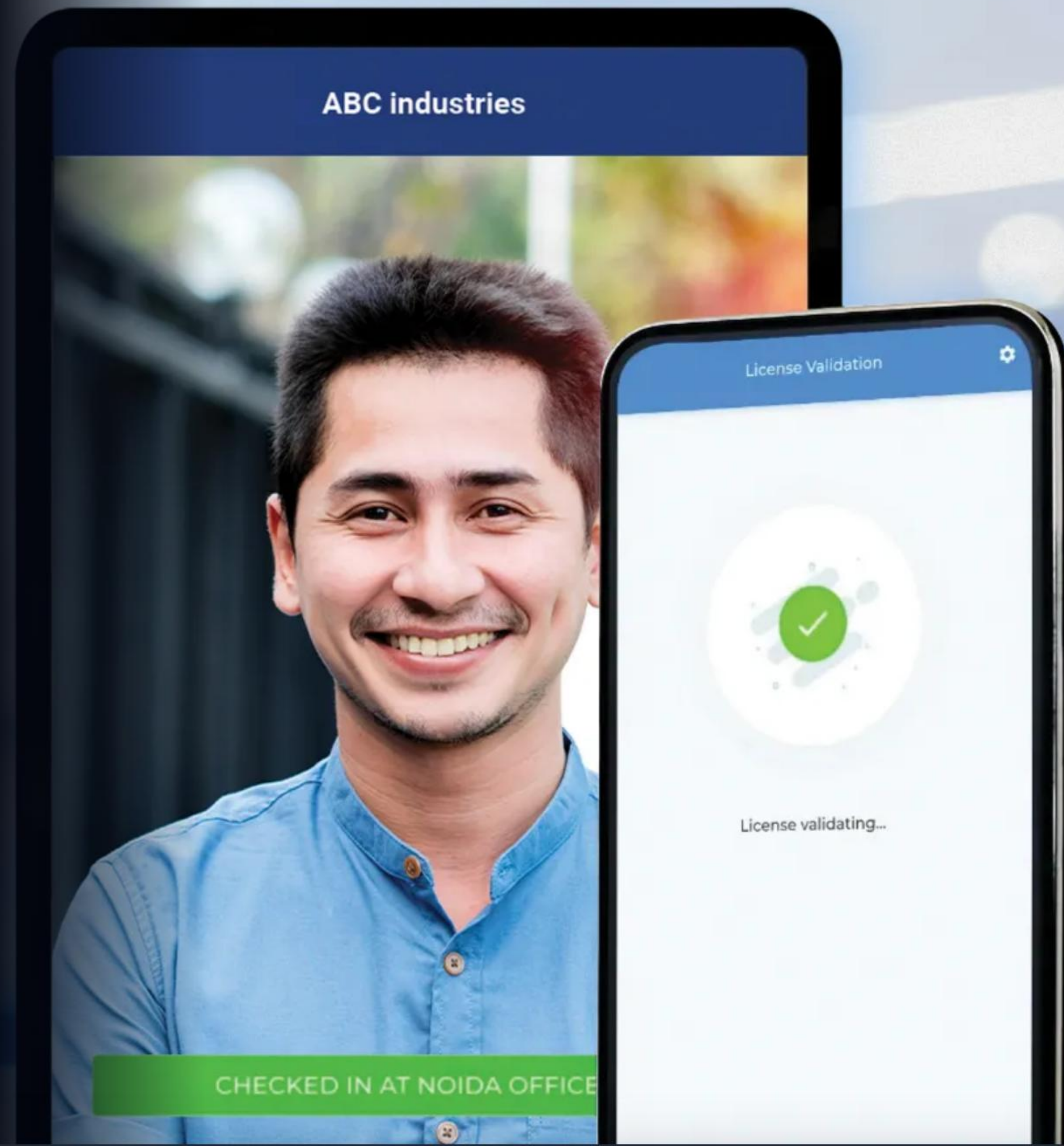
50% Semantic Match + 30% Skill Overlap + 20% Experience.

| Biometric Attendance

MobileFaceNet

Integrated lightweight neural network for real-time face verification on edge devices.

- < 100ms Inference Time
- Liveness Detection
- Geo-fencing support



| API Integration & Data Flow



Axios Interceptors

Centralized handling of request headers, token attachment, and global error handling (toasts).



TanStack Query

Manages server state with caching, optimistic updates, and background refetching for a snappy UX.



Real-time Sockets

Instant notifications for interview schedules, payroll generation, and approval requests.

| Journey: Smart Hiring



Post Job

HR Creates Listing



Sync

Push to LinkedIn



AI Screen

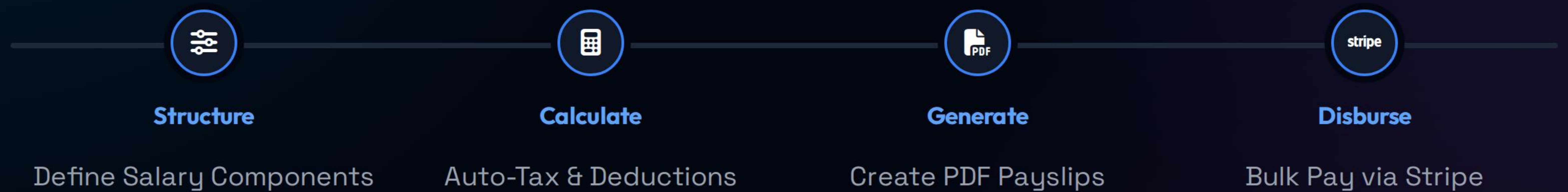
Auto-Rank Resumes



Onboard

Convert to Employee

| Journey: Payroll Processing



| Security & Scalability

Security First

 **CSRF & CORS:** Strict origin policies.

 **Encryption:** Data at rest and in transit.

 **Audit Logs:** Track every sensitive action.

Built to Scale

 **Dockerized:** Consistent deployment everywhere.

 **Caching:** Redis for high-speed data retrieval.

 **Load Ready:** Stateless API design.

| Future Roadmap



Mobile App

Native React Native application for on-the-go access.



Advanced Analytics

Predictive insights for employee retention and budget forecasting.



Multi-Language

i18n support for global enterprise deployment.

Q&A

Thank you for your attention.



/nexhr-project



contact@nexhr.tech

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